

ADR-COM-15-HIB MI Data Views.

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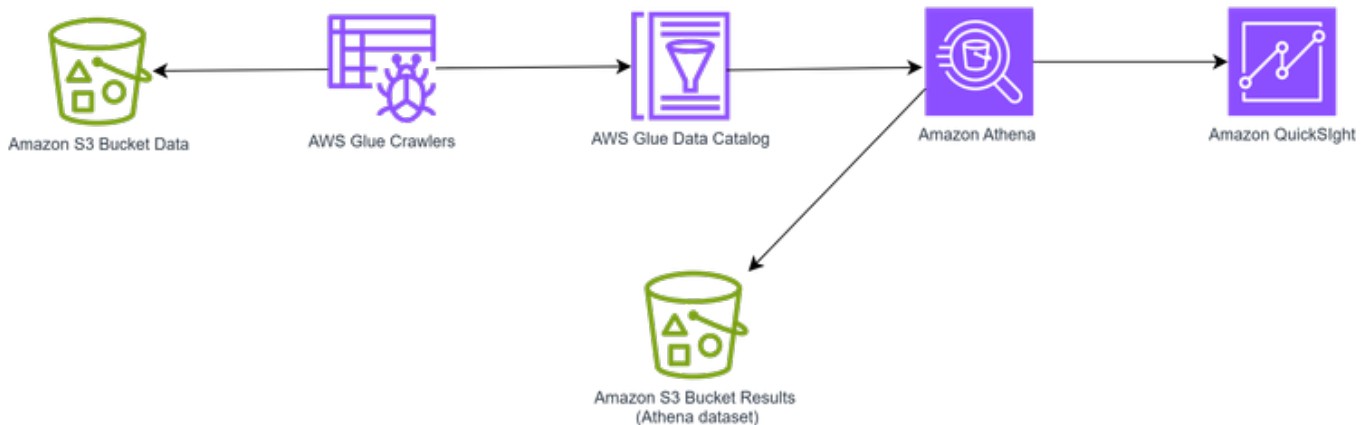
Deploying QuickSight Dashboards Across Multiple AWS Accounts Using Athena Data Sources, Multi-Account User Setup, and Embedded Dashboards

Context

The ~~XXXX~~ organisation plans to deploy Amazon QuickSight dashboards across **200 AWS accounts** to address client-specific reporting requirements at a regular cadence. These dashboards are expected to leverage AWS member accounts client queryable structured datasets via **AWS Athena**. The deployment strategy must consider scalability, user access configuration, management, embedding of dashboards, and overall cost-effectiveness.

Architecture Schematic

Solution Diagram



Key factors influencing the architecture decision include:

A. Data Sources

- **Athena Data Sources:**
 - Clients with structured and queryable datasets already accessible through AWS Athena can leverage direct querying for enhanced data analysis and reporting.

B. Multi-Account User Management

- The project spans **200 AWS accounts**, each representing a distinct client. It is proposed that each account have:
 - **5 Readers** for dashboard access.
 - A centralized account with **3 Authors** for baseline dashboard creation and updates.
- The lack of CloudFormation (CF) support for QuickSight user setup adds significant administrative overhead in provisioning users and managing subscriptions across accounts.

C. Reader Capabilities

- The reader will be able to do the following:
 - Consume data stories, dashboards, and reports.
 - Interrogate questions of data with Amazon Q.

D. Dashboard deployment at scale

- **Key Considerations for Deploying QuickSight Dashboards at Scale**
 - There are several aspects to consider:
 - **Prerequisites and Template Parameters**
 - **Athena DataSource**: Each target account might have a unique Athena database and table to reference.
 - **CloudFormation** templates will be configured to accept parameters that can be customized per account. These parameters will be provided during the deployment process.
 - **Parameters in CloudFormation Template**:
 - **Athena Data Source**: If using Athena, (template.json)
 - **CloudFormation Stack for QuickSight Dashboards**
 - **Dashboard**: The template.json will define the visualizations and dataset structure for the QuickSight dashboard.
 - **IAM Roles**: Ensure IAM roles are set up to allow QuickSight in the target accounts to access the necessary data sources (S3, Athena).

E. Dashboard Embedding

- Clients will also require dashboards embedded in URLs for seamless access via the **Kolizeo Hub**.
- Embedding must enforce strict **row-level security (RLS)** to ensure users access only authorised data.

F. Security and Compliance

- Datasets must remain within client-owned accounts to satisfy data sovereignty and compliance requirements, especially within **Kolizeo's** highly regulatory confines.
- Cross-account access must use robust IAM roles and S3 bucket policies to ensure fine-grained control and encryption.

G. Cost Management

- The architecture must balance:
 - Author subscription costs for managing dashboards.
 - Reader subscription costs, especially given the scale of 1000 Readers across 200 accounts.
 - Embedding API usage and associated infrastructure costs.

H. Administrative Overhead

- QuickSight currently does not support user provisioning via CloudFormation or the use of other Infrastructure as Code (IaC) like terraform, requiring manual setup and subscription management for each account.
- This presents significant challenges in terms of time, scalability, and consistency.

Decision

The intention is to implement a scalable, dual-option QuickSight deployment architecture supporting Athena data sources by leveraging AWS cloud native services and resources where possible. Key decisions are outlined below:

A. Dashboard Template Deployment

1. **Source Account**:
 - a. Dashboards will be created in a centralized source QuickSight account by the 3 Authors.
 - b. Exported as reusable templates using the StartAssetBundleExportJob API.
 - c. Templates will be stored in a version-controlled S3 bucket (quicksight-template-storage) for cross-account access.
2. **Target Accounts**:
 - a. Templates will be imported into each target QuickSight account using the StartAssetBundleImportJobAPI.
 - b. Each dashboard will be linked to the client-specific dataset stored in the respective S3 bucket or Athena.

B. Multi-Account QuickSight User Setup

1. **Centralized Authors**:
 - 3 Authors in the centralized source account will handle dashboard creation, updates, and sharing.
2. **Distributed Readers**:
 - Each of the 200 target accounts will have **5 Readers** who can view dashboards.
 - Readers will access dashboards via their usual AWS accounts or by means of embedded URLs with security contexts ensuring appropriate access control.

C. Dashboard Deployment at scale

1. **Prerequisites:** IAM roles, cross-account access, tracking versions, and automating the deployment process.
2. **Caching strategy:** Import Mode: SPICE (Super-fast, Parallel, In-memory Calculation Engine), for performant processing.
3. **Source Account Setup**
 - a. **Dashboard Template:** Create and save your QuickSight dashboard as a baseline template. Export as QuickSight dashboard template JSON (Asset Bundle) and maintain ARN.
 - b. **Service Role for QuickSight:** Grant QuickSight permissions to access necessary resources.
 - c. **Export Permissions:** Use the StartAssetBundleExportJob API to export the dashboard template.
4. **Target Accounts Setup**
 - a. **AWS QuickSight Setup**
 - i. Enable and configure AWS QuickSight Enterprise Edition in all target AWS accounts.
 - ii. Configure cross-account access permissions between source and target accounts.
 - b. **Service Role for QuickSight:** Set up an IAM role in each target account to allow QuickSight access.
 - c. **IAM Roles and Permissions**
 - i. **Cross-account access:** Create IAM role in each target AWS account to allow QuickSight in the central account to access data sources and dashboards.
 - ii. **Permissions for QuickSight:** Configure necessary service permissions.
 - iii. **Athena permissions:** Configure IAM role for QuickSight to query Athena and access S3 query results.
 - iv. **S3 Permissions:** Set up cross-account S3 bucket policies for central account dashboard access.
 - d. **Data Sources (Athena) Setup**
 - i. Configure Athena in target accounts with necessary tables and ensure S3 data accessibility.
 - ii. **Athena/Glue Integration:** Set up datasets in Athena and configure Glue crawlers for metadata.
 - e. **Data Sources (S3 Bucket) Setup**
 - f. **Customize Template:**
 - i. Create parameterized CloudFormation template for QuickSight resources (datasets, data sources, dashboards).
 - ii. Define parameters for account-specific values (Athena databases, tables).
 - iii. Configure QuickSightRole with required permissions.
 - g. **Baselining Template:**
 - i. Store template.json in member AWS account data source and reference in CloudFormation for Amazon Athena data sources ensuring standardized dashboard layouts.
 - h. **Athena Query Execution:**
 - i. Configure Athena with required tables and queries for QuickSight access.
 - i. **Versioning**
 - i. **Dashboard Names:** Use semantic versioning (e.g., Dashboard-v1, Dashboard-v2) in CloudFormation template, use of a property e.g., version description
 - ii. **Templates:** Maintain separate CloudFormation templates per major version or use template versioning.
 - iii. CloudFormation template to incorporate metadata tracking using either S3 or DynamoDB.

D. Dashboard Embedding

1. **Embedding Approach:**
 - Embed dashboards using the GetDashboardEmbedUrl API for registered users.
 - Use GenerateEmbedUrlForAnonymousUser API where required for anonymous access.
2. **Row-Level Security:**
 - URLs will include user-specific context, enforcing RLS to restrict dashboard views to authorized datasets only.

E. Administrative Strategy

- Automate user provisioning and subscription setup using AWS CLI or SDK (boto3) scripts to mitigate the lack of CloudFormation support for QuickSight accounts.
- Track user and subscription provisioning status in a DynamoDB table for audit and troubleshooting purposes.

Consequences

Benefits

1. **Scalability:**
 - Enables seamless deployment across 200 AWS accounts with consistent architecture.
 - Supports Athena data sources to accommodate diverse client setups.
2. **Security:**
 - Enforces RLS to ensure data segmentation for multi-tenant scenarios.
 - S3 bucket policies and IAM roles provide fine-grained access control.
3. **User-Friendly Embedding:**

- Provides clients with intuitive URL-based access to dashboards without requiring direct QuickSight logins.
4. **Cost Optimization:**
- Unlimited Reader pricing (\$3/Reader/month) ensures predictable costs for frequent dashboard access.

Trade-offs

1. **High Administrative Overhead:**
 - Manual provisioning of QuickSight users and subscriptions increases operational complexity.
 - Managing 200 accounts with 1000 users requires significant coordination and scripting efforts.
2. **Embedding Challenges:**
 - Embedding API integration adds development effort and requires ongoing monitoring of security and performance.
3. **Limited Automation:**
 - Absence of CloudFormation support for QuickSight user setup prevents fully automated deployments, increasing the time to scale.
4. **Limited Customisation:**
 - There is limited ability to customise the dashboard as it is within an iframe on the frontend.  will not be able to apply its own CSS frame.
 - The suggested user (reader) capabilities will only allow users to consume data provided on the dashboards.

Cost Implications

Authors

- 3 Authors in a centralized account at **\$24/ or \$18 month** (if paid annually)*
- **Monthly Cost:** $3 \times 24 = 72 \text{ USD/month}$ $3 \times 24 = 72 \text{ USD/month}$
- **Annual Cost:** $72 \times 12 = 864 \text{ USD/year}$ $72 \times 12 = 864 \text{ USD/year}$
- ***Annual Cost (paid annually):** $3 \times 18 \times 12 = 648 \text{ USD/year}$

Readers

- 1000 Readers (5 Readers $\times$ 200 accounts) at \$3/month each
- **Monthly Cost:** $1000 \times 3 = 3,000 \text{ USD/month}$ $1000 \times 3 = 3,000 \text{ USD/month}$
- **Annual Cost:** $3,000 \times 12 = 36,000 \text{ USD/year}$ $3,000 \times 12 = 36,000 \text{ USD/year}$

Total Deployment Cost

1. **Monthly:** $72 + 3,000 = 3,072 \text{ USD/month}$
2. **Annual:** $864 + 36,000 = 36,864 \text{ USD/year}$

Notes:

1. To estimate the costs associated with QuickSight, an approximation of the quantity of authors and readers among  clients is employed.
2. QuickSight Capacity Pricing (Enterprise Edition with Capacity) is not being considered for the cost estimation.
3. Generative BI capabilities with Amazon Q in QuickSight is not being considered for the cost estimation.

Alternatives Considered

A. Centralized QuickSight Account

- **Description:**
 - Use a single QuickSight account for all dashboards and datasets.
- **Pros:**
 - Simplifies user management and reduces administrative overhead.
- **Cons:**
 - Violates data sovereignty requirements by centralizing datasets in one account.
 - Limits scalability due to dataset and resource constraints in QuickSight.

B. Fully Manual Setup

- **Description:**
 - Create accounts, users, and subscriptions manually for each client.
- **Pros:**
 - Requires no additional development effort.

- **Cons:**
 - Extremely time-intensive and prone to errors.

C. Automation via Custom Scripts

- **Description:**
 - Use AWS CLI or SDK to automate user and subscription provisioning.
- **Pros:**
 - Reduces manual effort while providing flexibility.
- **Cons:**
 - Requires development and maintenance of scripts.

Related Decisions

- To be updated

Reference Sources

1. <https://aws.amazon.com/solutions/guidance/multi-account-environments-on-amazon-quicksight/>
2. <https://aws.amazon.com/blogs/business-intelligence/automate-and-accelerate-your-amazon-quicksight-asset-deployments-using-the-new-apis/>
3. <https://aws.amazon.com/blogs/big-data/migrate-amazon-quicksight-across-aws-accounts/>

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